

# Ruthenium Brachytherapy for Uveal Melanoma, 1979–2003

## Survival and Functional Outcomes in the Swedish Population

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**Purpose:** To evaluate observed and relative survival rates, enucleation rates, and visual outcome after ruthenium 106 brachytherapy for uveal melanoma.

**Design:** Retrospective cases series from the Swedish national referral center.

**Participants:** Five hundred seventy-nine patients (579 eyes) with choroidal or ciliary body melanomas, including 55 tumors more than 7 mm in height, treated with ruthenium episcleral plaques from January, 1979, through April, 2003.

**Methods:** Clinical and radiotherapy data were extracted from a dedicated database, and survival status was determined through population registries. Tumor size was classified according to the Collaborative Ocular Melanoma Study criteria. The 5- and 10-year relative survival rates were estimated, and univariate and multivariate regression models were constructed for predictive factors on observed survival, enucleation, and visual deterioration.

**Main Outcome Measures:** Observed and relative survival rate, proportion of secondary enucleation, deterioration of visual acuity to less than 0.5, respectively, to 0.1 or worse.

**Results:** Tumors were classified as small in 10.5%, medium in 78.4%, and large in 9.2% of patients. The 5- and 10-year observed overall survival rates were 83.3% and 71.5%, respectively, and the corresponding relative rates were 95.5% and 94%, respectively. Factors predicting survival were tumor diameter, patient age, and secondary enucleation. One hundred six patients (18%) underwent enucleation up to 14 years after plaque treatment. The only predictive factor for enucleation was tumor size. At 5 years, 31% of the patients retained 0.5 visual acuity or better, and 49% retained better than 0.1 visual acuity. Predictive factors for visual deterioration were visual acuity and distance from posterior tumor border to the foveola.

**Conclusions:** After ruthenium brachytherapy for uveal melanoma, the survival rates and visual outcomes in this population-based investigation were similar to previously published results. The eye was retained in 81.7% of patients. Careful patient selection (presently we only treat melanomas 7 mm or smaller in height) and life-long monitoring for recurrences is warranted. *Ophthalmology* 2005;112:834–840 © 2005 by the American Academy of Ophthalmology.

Uveal melanoma threatens both the patient's life and visual function. Treatment methods such as enucleation,<sup>1,2</sup> brachytherapy,<sup>3–6</sup> charged particle irradiation,<sup>7–9</sup> and local resection<sup>10,11</sup> seem to have similar survival outcomes.<sup>12–14</sup> Most important is to prevent death resulting from uveal melanoma, but both patients and ophthalmologists also focus on the preservation of visual function, cosmetic appearance, and quality of life.<sup>15</sup>

Since Lommatzsch<sup>16</sup> introduced ruthenium brachytherapy for uveal melanoma in the 1960s, many institutions in Europe have gained experience using this isotope for treating small to medium melanomas.<sup>17–21</sup> In North America, the iodine 125 applicator has become the primary choice for brachytherapy, and this radionuclide therefore was chosen in the brachytherapy arm of the Collaborative Ocular Mel-

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anoma Study (COMS) when investigating medium-sized choroidal melanomas. With a growing bulk of evidence from results of eye-sparing treatments suggesting a similar survival outcome, the advantages and disadvantages between different approaches to treatment are now of greater concern. In this paper, we present the results of 24 years of experience with brachytherapy for uveal melanoma, the study base being the entire Swedish population, and the patients were treated at a single center with ruthenium 106 ( $\text{Ru}^{106}$ ).

## Patients and Methods

### Patients

Ruthenium brachytherapy was introduced in 1979 in Sweden as a treatment method for posterior uveal melanomas. In the Swedish health care system, because of the relatively small population (9 million inhabitants in 2004), the practice of ocular oncology has been centralized to one national referral center at the Department of Ophthalmology at Karolinska Hospital, which in February 1990 was relocated to St. Erik's Eye Hospital.

Eligible for the present study were all patients who received ruthenium brachytherapy for a choroidal or ciliary body melanoma from January 1, 1979, through April 30, 2003. The indication for treatment was mainly small to medium uveal melanomas, although some large melanomas were treated at the patient's request or if the fellow eye lacked useful vision. Until 1999, when iodine plaques became the primary choice for the large melanomas selected for brachytherapy, ruthenium was the sole isotope in use for episcleral brachytherapy. In addition, 20 patients underwent charged particle irradiation with protons from 1989 through 1991 as an alternative to enucleation or brachytherapy. Data from a total of 579 patients (278 men, 301 women) were retrieved from a database (Medlog software; Information Analysis Corporation, Incline Village, NV), set up in the early 1980s at the Department of Oncology at Karolinska Hospital to monitor all Swedish patients receiving ocular radiotherapy. Because of strict centralization of brachytherapy and with only one institution providing appropriate dosimetry of ruthenium plaques (Department of Hospital Physics, Karolinska Hospital), an almost complete inclusion rate is assumed, with the exception of the rare patient who might have chosen treatment outside the country and, potentially, patients with clinically overt metastases at diagnosis who declined brachytherapy when suggested as palliative treatment. Clinical and radiotherapy data were extracted from this database along with records from hospital files. The follow-up period extended to November 1, 2003, thus allowing for a minimum of 6 months of follow-up regarding clinical data.

Because of the unique personal national registration number, which is cross-linked to the continuously updated Swedish Population Registry, each individual could be followed up regarding their vital status on April 30, 2004, which allowed for a minimum of 12 months of follow-up on survival. The study was approved by the Regional Research Ethical Committee in Stockholm in accordance with the tenets of the World Medical Association's Declaration of Helsinki.

### Tumors

The tumors were diagnosed clinically as choroidal or ciliary body melanomas. To evaluate the diagnostic accuracy of suspected uveal melanomas, one of the authors (SS, unpublished data) reexamined a random sample of 916 paraffin-embedded specimens

originally reported as uveal melanoma from 1960 through 1998. In only 3 cases (0.33%) did the original diagnosis require alteration.

Before treatment, the clinical work-up included best-corrected visual acuity using a decimal scale, biomicroscopy, indirect and direct ophthalmoscopy, transillumination, ultrasonography with standardized A and B scan, and fundus photography. In some instances, fluorescein angiography was undertaken. Fine-needle aspiration biopsies were not used.

The location (posterior to equator, equator up to pars plana, ciliary body), measurements (in millimeters) of the largest basal diameter, tumor apical height, and distance from the posterior border of the tumor to the foveola and to the optic disc were recorded. The tumor size was classified according to the COMS criteria. The small tumors measured less than 2.5 mm in apical height and 5 mm or more in longest basal diameter, the medium tumors ranged from 2.5 to 10 mm in apical height and 16 mm or less in longest basal diameter. The tumors classified as large had a longest basal diameter of more than 16 mm and height of more than 2 mm, or an apical height of more than 10 mm or, in the case of tumor growth within 2 mm from the optic disc, an apical height of more than 8 mm. Unlike in the COMS, a juxtapapillary tumor was coded as large if the mass encompassed more than 90° of the optic disc circumference.

Before selection for treatment, patients underwent a general medical examination to detect metastases resulting from uveal melanoma. The investigation was made in collaboration with internists at the referring hospitals and, depending on the local facilities, a chest radiograph, liver ultrasonography, or abdominal computerized tomography scan was undertaken, along with an analysis of liver function enzymes.

### Treatment

The criteria for ruthenium brachytherapy were mainly medium melanomas, or small melanomas with documented growth. The only isotope used in this study was  $\text{Ru}^{106}$ , and the applicators were standard episcleral plaques with a diameter of 11.6 to 20.2 mm (CCA, CCB, CCX, CCZ, COB, or CIB) manufactured by Bebig GmbH (Berlin, Germany). Retreatment, when indicated, was undertaken with a ruthenium plaque only, or in combination with argon laser photocoagulation, which in 1998 was replaced by transpupillary thermotherapy.

The applicators were replaced continuously every year, corresponding to the half-life (368 days) for the  $\text{Ru}^{106}$  isotope. For each new applicator, the dose rate was measured with a p-type silicon detector developed at the Department of Hospital Physics at Karolinska Hospital<sup>22</sup> estimated to give an accuracy of  $\pm 5\%$ . The radiotherapy data recorded were the dose (Gy) and dose rate (Gy/hour) at the tumor apex and the tumor apex + 1 mm and at the scleral surface. The target dose to the tumor apex (prescription point) was calculated with an additional safety margin of 1 mm, apart from the 1 mm for sclera (apex + 1 mm) and set at 100 Gy. Consequently, the dose to the actual tumor apex exceeded 100 Gy in many instances. The dose was reduced in a few cases to not exceed the limit set for the scleral dose (1000 Gy, in 1986 increased to 1500 Gy).

### Follow-up

Regular follow-up was scheduled at 3, 6, and 12 months after therapy, and then annually or semiannually if local control had been achieved. At each visit, best-corrected visual acuity, slit-lamp examination, biomicroscopy, indirect and direct ophthalmoscopy, transillumination, ultrasonography with standardized A and B scan, and fundus photography were undertaken. The visual acuity was measured on a decimal scale chart at a distance of 5 m

(Monoyer-Granström) ranging from 1.0 (corresponding to 20/20) to 0.1 (corresponding to 20/200). Visual acuity worse than 0.1 was recorded either as counting fingers, hand movements, or light perception.

If an inadequate response to brachytherapy was noted (progression after initial regression or, unlike the COMS, lack of regression), retreatment as indicated earlier was undertaken. Semiannual screening for metastases of uveal melanoma was performed in accordance with the wishes of the patient and in collaboration with an internist at the referring hospital. Screening was performed by serum liver enzymes, liver ultrasonography, or both.

## Statistical Methods

Relative survival was analyzed by the life table method described by Hakulinen and Tenkanen<sup>23</sup> in which 1 – the relative survival rate is an estimate of the fraction of deaths caused by the studied disease. The relative survival rate is defined as the ratio of the survival rate observed in a patient group of interest to the survival rate expected in a group of people from the general population, similar to the patient group with respect to the factors affecting survival (patient age, gender, and calendar period). This method provides estimates for the all-cause mortality rates, which correspond to the observed rates, and the tumor-specific mortality represented as the relative rates. The Swedish population life tables, covering the ages 0 to 99 years by gender, were used in the calculations. Loss of visual acuity and enucleation were estimated by actuarial and life table methods. Visual acuity was analyzed with 2 end points, deterioration of visual acuity to less than 0.5 (<20/40) or to 0.1 or less ( $\leq$ 20/200). The visual acuity levels recorded at the annual visits were used in the analysis; consequently, patients with less than 1 year of follow-up were excluded before the first event (deterioration of visual acuity) took place. The loss to the level of interest was established at the time point when it was first recorded, if the following visit confirmed the reduction of visual acuity. Otherwise, the visual acuity at the last follow-up visit was accepted. If the patient had undergone enucleation with a preoperative visual acuity of more than 0.1, it was regarded as an event at next annual follow-up.

Cox regression was used for univariate and multivariate analyses on overall survival, visual outcome, and enucleation. Time to death from treatment was analyzed with enucleation as a time-dependent covariate in Cox regression, because enucleation took place after treatment start, when the follow-up with respect to death had already started.<sup>24</sup> The variables from the univariate model were entered with stepwise forward procedure into the multivariate analysis if  $P < 0.05$  to fit the final model. The relative hazards ratios with 95% confidence intervals and  $P$  values were estimated.

The variables were analyzed as categories except for age, tumor height, largest tumor diameter, distance from posterior tumor border to the optic disc, distance from posterior tumor border to the foveola, radiation dose rates, and radiation doses. The calculations were made with SPSS<sup>25</sup> and a computer program package for relative survival analysis.<sup>26</sup>

## Results

During the investigation period, 579 Swedish patients underwent ruthenium brachytherapy for uveal melanoma. The patient demographics, tumor characteristics, and radiotherapy data are given in Table 1. Two patients had bilateral uveal melanoma, but only the eye first treated with brachytherapy was entered into the analysis. The patients had a median age of 63 years at treatment (range, 17–93 years). The median largest tumor diameter was 10 mm

Table 1. Baseline Patient and Tumor Characteristics at Diagnosis and Radiotherapy Data on 579 Patients Treated with Ruthenium Brachytherapy for Uveal Melanoma\*

|                                                                 | Ruthenium Brachytherapy (n = 579) |      | Medium-Sized Melanomas (n = 454) |      |
|-----------------------------------------------------------------|-----------------------------------|------|----------------------------------|------|
|                                                                 | n                                 | %    | n                                | %    |
| Age (yrs)                                                       |                                   |      |                                  |      |
| <45                                                             | 65                                | 11.2 | 47                               | 10.4 |
| 45–59                                                           | 165                               | 28.5 | 123                              | 27.1 |
| 60–74                                                           | 234                               | 40.4 | 194                              | 42.7 |
| $\geq 75$                                                       | 115                               | 19.9 | 90                               | 19.8 |
| Gender                                                          |                                   |      |                                  |      |
| Male                                                            | 278                               | 48.0 | 213                              | 46.9 |
| Female                                                          | 301                               | 52.0 | 241                              | 53.1 |
| Calendar period for initial treatment                           |                                   |      |                                  |      |
| 1979–1984                                                       | 32                                | 5.5  | 22                               | 4.8  |
| 1985–1989                                                       | 111                               | 19.2 | 86                               | 18.9 |
| 1990–1994                                                       | 107                               | 18.4 | 92                               | 20.3 |
| 1995–1999                                                       | 188                               | 32.4 | 148                              | 32.6 |
| 2000–2003                                                       | 141                               | 24.3 | 106                              | 23.3 |
| Best-corrected visual acuity at diagnosis                       |                                   |      |                                  |      |
| $\leq 0.1$ ( $\leq$ 20/200)                                     | 75                                | 13.0 | 55                               | 12.1 |
| 0.13–0.4 (20/160–20/50)                                         | 120                               | 20.2 | 90                               | 19.8 |
| 0.5–0.65 (20/40–20/30)                                          | 125                               | 21.6 | 102                              | 22.4 |
| 0.7–0.9 (>20/30–<20/20)                                         | 115                               | 19.9 | 88                               | 19.4 |
| $\geq 1.0$ ( $\geq$ 20/20)                                      | 140                               | 24.2 | 115                              | 25.3 |
| Missing value                                                   | 4                                 | 0.7  | 4                                | 0.9  |
| Tumor size (COMS criteria)                                      |                                   |      |                                  |      |
| Small                                                           | 61                                | 10.5 |                                  |      |
| Medium                                                          | 454                               | 78.4 | 454                              | 100  |
| Large                                                           | 53                                | 9.2  |                                  |      |
| Not classifiable                                                | 11                                | 1.9  |                                  |      |
| Largest basal diameter (mm)                                     |                                   |      |                                  |      |
| 3.0–7.9                                                         | 155                               | 26.8 | 110                              | 24.2 |
| 8.0–10.9                                                        | 204                               | 35.2 | 163                              | 35.9 |
| 11.0–16.0                                                       | 199                               | 34.4 | 172                              | 37.9 |
| 16.1–23.0                                                       | 11                                | 1.9  |                                  |      |
| Missing value                                                   | 10                                | 1.7  | 9                                | 2.0  |
| Apical height (mm)                                              |                                   |      |                                  |      |
| 1.0–2.4                                                         | 63                                | 10.9 |                                  |      |
| 2.5–5.0                                                         | 312                               | 53.9 | 281                              | 61.9 |
| 5.1–7.0                                                         | 149                               | 25.7 | 129                              | 28.4 |
| >7.0                                                            | 55                                | 9.5  | 44                               | 9.7  |
| Distance from posterior tumor border to foveola (mm)            |                                   |      |                                  |      |
| 0.0–2.0                                                         | 274                               | 47.3 | 208                              | 45.8 |
| 2.5–5.9                                                         | 161                               | 27.8 | 126                              | 27.8 |
| $\geq 6.0$                                                      | 103                               | 17.8 | 90                               | 19.8 |
| Missing value                                                   | 41                                | 7.1  | 30                               | 6.6  |
| Distance from posterior tumor border to edge of optic disc (mm) |                                   |      |                                  |      |
| 0.0–2.9                                                         | 214                               | 37.0 | 142                              | 31.3 |
| 3.0–5.9                                                         | 216                               | 37.3 | 182                              | 40.1 |
| $\geq 6.0$                                                      | 122                               | 21.1 | 112                              | 24.7 |
| Missing value                                                   | 27                                | 4.7  | 18                               | 4.0  |
| Location of posterior tumor border                              |                                   |      |                                  |      |
| Posterior to equator                                            | 533                               | 92.1 | 420                              | 92.5 |
| Anterior to equator                                             | 15                                | 2.6  | 11                               | 2.4  |
| Ciliary body melanoma                                           | 12                                | 2.1  | 12                               | 2.6  |
| Not classifiable                                                | 19                                | 3.3  | 11                               | 2.4  |
| Dose to prescription point (apex + 1 mm; Gy)                    |                                   |      |                                  |      |
| <80                                                             | 49                                | 8.4  | 38                               | 8.4  |
| 80–99                                                           | 20                                | 3.4  | 16                               | 35.2 |

(continues)

Table 1. (Continued.)

|                           | Ruthenium Brachytherapy (n = 579) |      | Medium-Sized Melanomas (n = 454) |      |
|---------------------------|-----------------------------------|------|----------------------------------|------|
|                           | n                                 | %    | n                                | %    |
| 100                       | 437                               | 75.4 | 348                              | 76.7 |
| >100                      | 16                                | 2.8  | 12                               | 2.6  |
| Missing value             | 57                                | 9.8  | 40                               | 8.8  |
| Scleral surface dose (Gy) |                                   |      |                                  |      |
| 0–499                     | 289                               | 49.9 | 207                              | 45.6 |
| 500–999                   | 182                               | 31.6 | 156                              | 34.4 |
| 1000–1499                 | 85                                | 14.7 | 72                               | 15.0 |
| ≥1500                     | 23                                | 4.0  | 19                               | 4.2  |

COMS = Collaborative Ocular Melanoma Study.

\*Data concerning the medium-sized melanomas (Collaborative Ocular Melanoma Study criteria; 454 patients) were extracted from this pool.

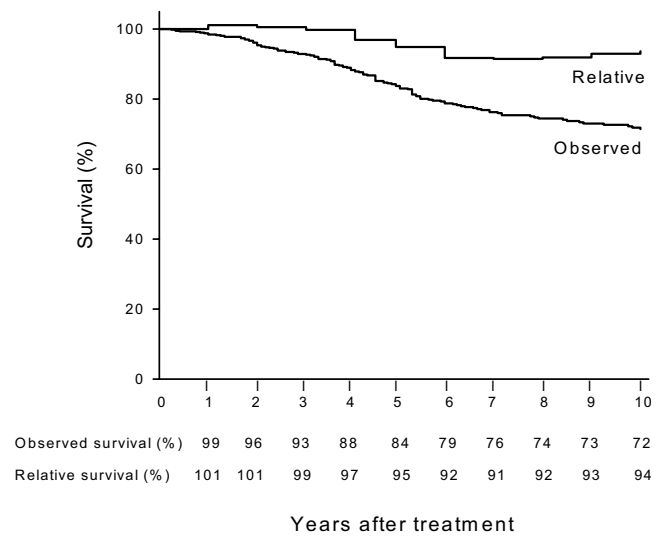
(range, 3–23 mm), and the median apical height was 4.3 mm (range, 1–13.1 mm). The median distance from the posterior border of the uveal melanoma to the foveola and the edge of the optic disc was 2.0 and 3.0 mm, respectively. The median best-corrected visual acuity at the time of diagnosis was 0.65. No patient had clinically detectable metastases before surgery or signs of extrascleral growth during surgery.

During the investigation period, 130 patients died and 2 patients were lost to follow-up because of emigration. The follow-up time for surviving patients ranged from 6 to 292 months, with a median of 82 months and a mean of 97 months. The follow-up period for the deceased patients varied from 3 to 232 months, with a median of 55 months and a mean of 67 months.

## Survival Analysis

During the first year after treatment, 7 patients died, of whom 5 had medium-sized melanomas. The observed (all-cause) survival rate was 98.9%, with a relative (tumor-specific) survival rate of 101%, which indicates no excess mortality in the uveal melanoma patient group during the first posttreatment year compared with the population in general. During the third posttreatment year, the relative survival rate decreased to less than 100%. The 5-year relative (tumor-specific) survival rate was 94.9% and was reduced further through the following 3 years to approximately 92%. Uveal melanoma mortality based on relative survival seemed to persist up to 9 years after diagnosis (Fig 1). The patients with medium melanomas featured a relative survival rate of 95.5% at 5 years and 92% for the following 3 years. The observed (overall) survival rate for the entire patient group at 5 years was 83.8% and was 71.5% at 10 years. The corresponding observed survival rates for patients with medium uveal melanomas were 84.2% (95% confidence interval, 80.4–87.8) at 5 years and 71.6% (95% confidence interval, 66.4–76.7) at 10 years.

In the univariate analyses, the factors predicting overall (observed) survival outcome were gender, patient age (in 10-year increment), tumor diameter, tumor height, and enucleation. In the final multivariate model, the significant predictive factors for overall patient survival were younger patient age ( $P < 0.001$ ), decreasing tumor diameter ( $P < 0.001$ ), and no enucleation ( $P = 0.003$ ). The overall survival function after enucleation (Fig 2) showed a somewhat higher rate of deaths through the first 5 years, which justifies the results from the Cox regression treating enucleation as a time-dependent variable.



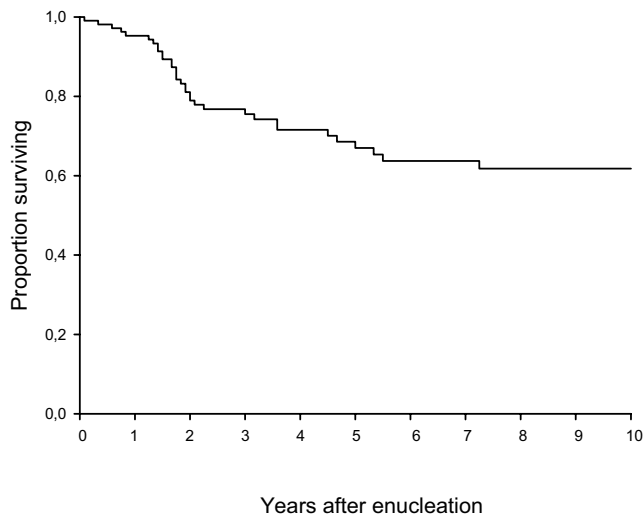
**Figure 1.** Observed (all-cause) and relative (tumor-specific) survival after ruthenium brachytherapy for uveal melanoma (579 patients).

## Analysis of Enucleation

One hundred six patients eventually underwent enucleation, with a 5-year cumulative incidence of 16.8%. Most enucleations ( $n = 80$ ) were performed within the first 3 years after plaque treatment: in 60 patients because of inadequate local tumor control and in 17 patients because of ocular side effects such as neovascular glaucoma or lack of useful vision. In 3 patients, the reason for enucleation could not be established from the records. Still, another 25 patients eventually lost their eye up to 10 years after initial treatment (Fig 3). One patient experienced a late relapse 14 years after treatment and underwent enucleation. The reason for enucleation in 79 patients (74.5%) was a lack of tumor regression or tumor progression after initial regression. In the case of insufficient local tumor control, retreatment with ruthenium brachytherapy was performed in 44 patients (7.6%), of whom 14 later underwent enucleation. The secondary plaque treatment brachytherapy was performed at 2 months to 6.5 years after primary brachytherapy. Of the 69 patients who received a dose to the prescription point of less than 100 Gy (so as not to exceed the maximum scleral dose), 16 patients (23.2%) underwent secondary enucleation, in 11 cases because of deficient local tumor control. Only 19.8% of the enucleations were the result of ocular side effects such as neovascular glaucoma or significantly impaired vision. The univariate factors predicting the risk of enucleation were tumor size (COMS classification), dose rate at the scleral surface, and apex dose. In the multivariate model, only increasing tumor size predicted later enucleation ( $P = 0.011$ ). Because controversy exists regarding whether ruthenium plaques should be used for uveal melanomas in the range of 5 to 7 mm in tumor height (with risk of not achieving an effective apex dose), we studied the proportion of enucleation in the patient group with a 3- to 5-mm tumor height ( $n = 270$ ) compared with the patients with a 5.1- to 7-mm tumor height ( $n = 149$ ) and found no statistically significant difference ( $P = 0.716$ ). Although the power to detect such a difference would be only 5%, under the assumption of a 2-sided hypothesis and a 5% risk level.

In the former group, 51 patients (18.9%) underwent enucleation; in the latter group, 26 patients (17.4%) underwent enucleation. However, because the distance to the optic disc is of importance in placing the plaque efficiently, it should be noted that the group with smaller tumor height had a statistically significantly shorter mean distance to the optic disc (3.7 mm) compared with the group with larger tumor height (4.8 mm;  $P = 0.02$ ).





**Figure 2.** Observed (all-cause) survival after secondary enucleation in patients treated with a ruthenium episcleral plaque for uveal melanoma.

### Analysis of Visual Outcome

At the time of diagnosis, 65.7% of the patients had a best-corrected visual acuity of 0.5 (20/40) or more and 13% had a visual acuity of 0.1 (20/200) or less. The overall decline in visual acuity after brachytherapy is shown in Table 2. Patients were withdrawn from further analysis if enucleation had been performed. At 5 years of follow-up, 64 of the remaining 171 patients (37.2%) had a visual acuity of 0.5 or better, but 67 patients (39.2%) had less than 0.1 vision. At 10 years of follow-up, 32 of 58 patients retained 0.13 or better vision. The proportion of the 326 patients with initial visual acuity 0.5 or better and with a follow-up of at least 1 year who retained at least 0.5 vision is shown in Figure 4.

In the univariate analyses, the factors predicting visual deterioration to below the level of 0.5 were: dose rate at tumor apex, dose rate at prescription point (apex + 1 mm), scleral surface dose, tumor height, tumor diameter, gender, initial visual acuity, and the distance from the tumor to the foveola. In the final multivariate Cox regression analysis, increasing tumor height ( $P = 0.001$ ), decreasing initial visual acuity ( $P = 0.001$ ), and decreasing distance from the tumor to the foveola ( $P = 0.021$ ) remained significant risk factors.

The visual outcome for the group of patients with an initial visual acuity of better than 0.1 and follow-up of 1 year or more is illustrated in Figure 5. The predictive factors for visual deterioration to 0.1 or worse in the univariate analyses were: distance from posterior tumor border to the optic disc, distance from the posterior tumor border to the foveola, and initial visual acuity. In the multivariate analysis, only decreasing distance from posterior tumor border to the foveola ( $P = 0.015$ ) and decreasing initial visual acuity ( $P = 0.001$ ) showed independent impact on visual loss to the level of 0.1 or worse.

### Discussion

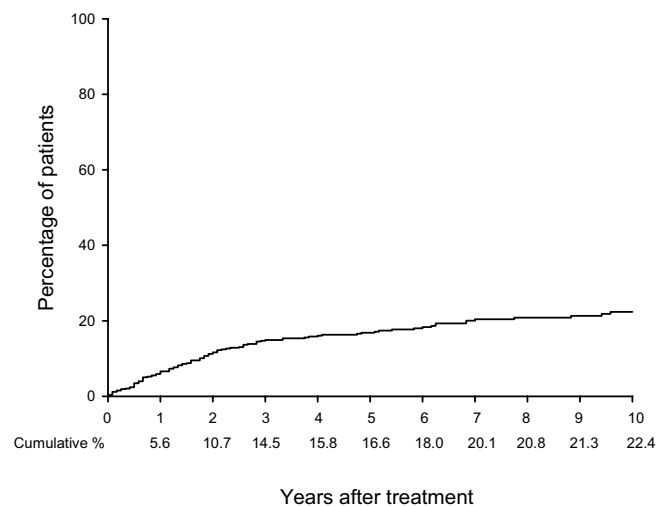
For the last decades, evidence has accumulated, including the prospective COMS,<sup>5</sup> indicating that the choice of treatment for uveal melanoma does not affect patient survival. However, the reported survival rates are not entirely comparable because of differences in patient selection, tumor characteristics, different

staging, and statistical approaches. In our series, 78% of tumors were medium (COMS classification), compared with the series of 309 uveal melanoma patients reported by Lommatzsch<sup>16</sup> in which 70% were T1 and 8% were T2 (TNM classification), with a 5-year overall survival rate of 84% and a melanoma-related survival rate of 89%. Rouberol et al<sup>17</sup> recently published results of 213 patients treated with ruthenium brachytherapy. Their patient group contained 19% T1 tumors, 36% T2 tumors, and 45% T3 tumors. The 5-year overall survival rate was 82%. We found an observed 5-year survival rate of 84% and a 5-year relative survival rate of 95%.

These data indicate that both the overall and tumor-specific survival outcomes we report largely are in accordance with previously published results. The reasons for using the relative survival method when estimating tumor-specific survival are 2-fold: this method accounts for the competing risks of death resulting from other diseases, and the method does not require knowledge of the underlying cause of death. In Sweden, we recently found that deaths due to uveal melanoma only had an accurate classification rate of 46% in death certificates and that autopsy rates are declining.<sup>27</sup>

In our series, the predictive factors for overall survival by multivariate analysis were patient age, tumor diameter, and subsequent enucleation. The reason enucleation emerged as an independent predictive factor could be that enucleated eyes harbor more aggressive tumors that metastasize early. However, removal of the tumor-containing eye also may promote the growth of distant metastases, and more knowledge about host-tumor interaction is required, as discussed recently by Singh et al.<sup>14</sup> Even if enucleation per se does not threaten the patient's life, the goal of preserving the eye is not achieved.

The overall eye salvage rate among our patients was 81.7%, with a 5-year cumulative incidence of enucleation of 16.8%. However, some patients treated during the first decade of this study would not be selected for this treatment today because other therapeutic options are available. The occurrence of late relapses after 5 years to more than 20 years after brachytherapy has been observed, both in series



**Figure 3.** Cumulative percentage of patients who underwent enucleation after ruthenium brachytherapy for uveal melanoma.

Table 2. Distribution of Visual Acuity after Ruthenium Brachytherapy

| Visual Acuity           | At Diagnosis<br>n (%) | 1 Year<br>n (%) | 2 Years<br>n (%) | 3 Years<br>n (%) | 5 Years<br>n (%) | 10 Years<br>n (%) |
|-------------------------|-----------------------|-----------------|------------------|------------------|------------------|-------------------|
| ≤0.1 (≤20/200)          | 75 (13.0)             | 105 (24.3)      | 109 (32.5)       | 103 (38.0)       | 67 (39.2)        | 26 (44.8)         |
| 0.13–0.4 (20/160–20/50) | 120 (20.2)            | 125 (28.9)      | 87 (26.0)        | 69 (25.5)        | 40 (23.4)        | 16 (27.6)         |
| 0.5–0.65 (20/40–20/30)  | 125 (21.6)            | 71 (16.4)       | 53 (6.3)         | 35 (12.9)        | 22 (12.9)        | 8 (13.8)          |
| 0.7–0.9 (>20/30–<20/20) | 115 (19.9)            | 68 (15.7)       | 39 (11.6)        | 34 (12.5)        | 24 (14.0)        | 4 (6.9)           |
| ≥1.0 (≥20/20)           | 140 (24.2)            | 63 (14.6)       | 47 (14.0)        | 30 (11.1)        | 18 (10.5)        | 4 (6.9)           |
| Missing value           | 4 (0.7)               |                 |                  |                  |                  |                   |
| Median visual acuity    | 0.65                  | 0.40            | 0.40             | 0.25             | 0.20             | 0.16              |
| Total patients          | 579                   | 432             | 335              | 271              | 171              | 58                |

using iodine isotopes<sup>28</sup> and ruthenium isotopes.<sup>29</sup> We found delayed recurrences after 5 years or more in 17 patients, of whom 4 patients were managed with a second plaque but 13 patients required enucleation. In contrast to the COMS, lack of regression was an indication for enucleation in our patients, which may contribute to different 5-year cumulative risks of enucleation (12.5% in the COMS vs. 16.8% in our series).

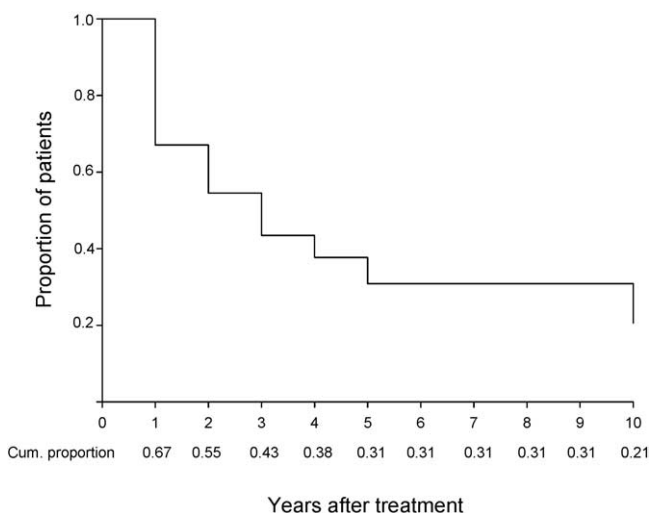
The reasons for enucleation in 20% of patients were ocular side effects such as pain, neovascular glaucoma, or an unsightly eye, which differs from the COMS,<sup>30</sup> in which 40% of the secondary enucleations in the iodine arm were the result of pain or low visual acuity. Our local complication rate seems to be in accordance with other ruthenium series in which 15% to 30%<sup>16,31</sup> of the enucleations were caused by ocular side effects.

The only independent factor for enucleation was tumor size, because dose rate at the scleral surface and tumor apex dose did not prove to be statistically significant by multivariate analysis. It could be argued that tumors with a height of more than 7 mm (55 patients) or tumors touching the optic disc (43 patients) may not have received an effective dose.

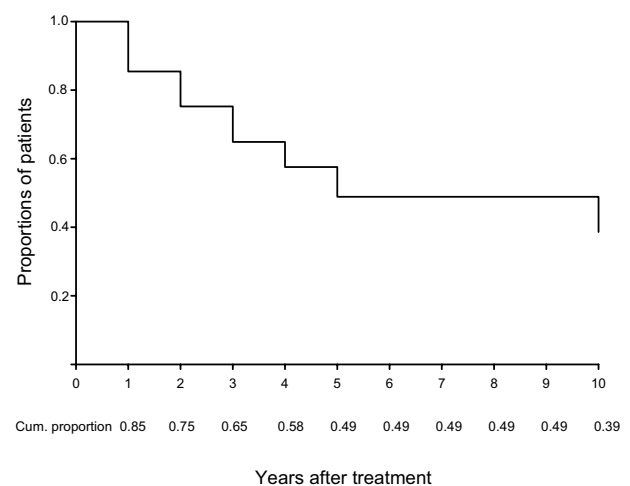
Because visual outcome for obvious reasons depends on

the relationship of the tumor border to the macula and optic disc, the proportion of patients retaining certain levels of visual acuity varies between case series. Apart from patient selection, the properties of the isotope used and the radiation dose differ. A direct comparison between our retrospective case series and the COMS<sup>5,30,32</sup> results with iodine brachytherapy for medium uveal melanomas therefore is not appropriate. Melanomas classified as medium-sized and treated with a ruthenium plaque were only in 44 cases (9.7%) of height greater than 7 mm, compared with 14% in the COMS. Also, dosimetric differences between applied treatment within the COMS and ruthenium brachytherapy as delivered in our case series could account for effects on tumor relapse rate and visual deterioration.

In our cohort, most tumors (92%) were located with their center posterior to the equator; in 47%, the posterior tumor border was within 2 mm of the foveola, and in 37%, within 2 mm of the optic disc. The only isotope used was ruthenium, and the dose at the prescription point was consistently aimed at 100 Gy with no variation over calendar periods. In 75% of tumors, the prescribed dose was achieved. Despite the posterior location of the uveal melanomas in our series, 5 years after brachytherapy 31% of patients who were still alive and had not undergone enucleation had a visual acuity of 0.5 or better, and 49% had a visual acuity of better than



**Figure 4.** Proportion of patients with an initial visual acuity of 0.5 or better and retaining 0.5 visual acuity or better after ruthenium brachytherapy for uveal melanoma. Initially, 326 patients (with at least 1 year of follow-up) were at risk.



**Figure 5.** Proportion of patients with an initial visual acuity of better than 0.1 and retaining this level of visual acuity after ruthenium brachytherapy for uveal melanoma. Initially, 404 patients (with at least 1 year of follow-up) were at risk.

0.1. Factors predicting the visual outcome for both the 0.5 and 0.1 levels proved to be initial visual acuity and the distance from the posterior tumor border to the macula.

Since ruthenium brachytherapy for uveal melanomas became available in Sweden 25 years ago, substantial experience has accumulated. We and others have found that the melanoma mortality rate is little affected by choice of isotope or if the patient undergoes primary enucleation. The chance of retaining the eye seems to be approximately 10% to 20%. With ruthenium, the local relapse rate (including lack of regression) is probably somewhat higher than with iodine, but complications such as neovascular glaucoma are less frequent. At 5 and 10 years after ruthenium brachytherapy, 49% and 39%, respectively, of eyes can still be expected to retain a visual acuity better than 0.1.

With current knowledge, we presently select only patients with a tumor height of less than 7 mm for ruthenium brachytherapy and frequently apply transpupillary thermotherapy as an adjunct if an edge recurrence appears. Theoretically, because data are lacking regarding whether an early detection of recurrence prolongs survival, the real need to monitor the patient closely is not proven. But in the absence of such data, and with recurrences appearing more than 10 years after treatment, we presently prefer to observe these patients, probably for decades.

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